

Examining Usability Issues in Blockchain-Based Cryptocurrency Wallets

Md Moniruzzaman, Farida Chowdhury, and Md Sadek Ferdous

Shahjalal University of Science and Technology, Sylhet, Bangladesh
monirulhasann@gmail.com, {farida-cse,sadek-cse}@sust.edu

Abstract. Blockchain has emerged as a revolutionary technology that has been envisioned to disrupt several industries, including financial domains with its decentralised and highly-secure design. However, since the beginning of its evolution, it has been highly criticised for the difficulties in dealing with it properly. As cryptocurrency is the most successful application of blockchain, we aim to identify the potential obstacles and usability issues, which might hinder its wide-scale adoption, of five applications (i.e., wallets) that are used to manage cryptocurrencies. Applying the analytical cognitive walk-through usability inspection method, we investigate common usability issues with desktop and mobile-based wallets. Our results reveal that both wallets lack good usability in performing the the fundamental tasks which can be improved significantly. We summarise our findings and point out the aspects where the issues exists so that improving those areas can result in better user experience and adoption.

Keywords: Usability, Human Computer Interaction, Cryptocurrency, Blockchain, Cognitive walk-through, User-interface, User-experience

1 Introduction

With the inception of Bitcoin in 2008 [1] a new form of digital currency called cryptocurrency has come into existence. The primary goal of such a digital currency is to carry out financial transactions among respective parties without any trusted intermediaries. This is fundamentally different to the existing financial transactions systems which rely on a single or group of trusted organisations for any financial transaction to happen. This strong property of a cryptocurrency is often regarded as a fundamental breakthrough in financial technology domains.

A cryptocurrency is underpinned by this novel technology called Blockchain which is an ordered data structure consisting of consecutive blocks of transactions linked together with a strict set of rules. These chained blocks are then distributed and stored by the nodes in a peer-to-peer network where each new block of transactions is created and appended at a predefined interval in a decentralised fashion by means of a consensus algorithm.

Blockchain exhibits several sought-after properties such as distributed consensus and the immutability and irreversibly of the blockchain state. It is even

envisioned that blockchain technology will effectively disrupt several existing application domains and there have been huge interests among the practitioners to explore this avenue. However, as of today, cryptocurrency has remained the most successful application of blockchain technology and because of the world-wide popularity, Bitcoin has become its poster-boy. Driven mostly by speculation and a hype-cycle, the price of a single Bitcoin even reached 19,783USD in December, 2017 [2]. Following the footsteps of Bitcoin, there are other cryptocurrencies which serve different purposes and target different applications domains. However, the common undertone among all these digital currencies is to either complement or supplement the fiat-currency based traditional monetary systems in different application scenarios. This can only be possible with a wide-scale adoption of these cryptocurrencies.

One of the crucial barriers for the wide-scale adoption of any novel technology is the issues in usability. Usability is the degree to which a software is easy to use and a good fit for the audience. A cryptocurrency heavily relies on cryptographic mechanisms which are often difficult to utilise in an effective manner, even for experienced security programmers [3]. Therefore, it is imperative to investigate how a general user can effectively use a cryptocurrency.

A cryptocurrency use asymmetric cryptography for security through a pair of keys: the private and public. A private key is a secret number that allows a user to use the currency. It must remain secret at all times as revealing it to third parties is equivalent to giving them control over the corresponding currency. As such, these keys are fundamental to manage and transfer the ownership of a cryptocurrency to another user. To facilitate the management of such keys, a software called *Wallet* is used to create and store the required cryptographic keys and hence, plays a vital role in all use cases of a cryptocurrency. If a user loses a wallet due to theft, hard disk crashes or malware, the user loses all the associated keys and hence, the cryptocurrencies. Unfortunately, if this happens, there is no way to recover the lost currencies. Another concern is that these keys are often encrypted with a password. If the user forgets the associated password, there is no way to decrypt and recover the private keys. Hence, there is no other recovery mechanism, provided that the backup is not taken. A wallet is essentially the interface by which a user interacts with the associated cryptocurrency and the underlying blockchain. Therefore, its usability must be studied to find out any usability issues in those wallets, which might be the key obstacles behind any global adoption.

Towards this aim, there have been a few studies, elaborated in the subsequent section. Unfortunately, all these studies focused only on Bitcoin. There are other popular cryptocurrencies which were mostly overlooked in those studies. We tackle this issue in this paper by presenting a usability study, based on a popular analytical method called *Cognitive Walkthrough*, of three popular cryptocurrencies, namely Bitcoin [1], Ethereum [4] and Ripple [5], and their different popular wallets. We provide a short background of these cryptocurrencies and their wallets in Section 2 along with a discussion of different usability study

methods. Then, in Section 3, we analyse the existing works in relation to our current papers.

For each of the selected wallets, recruited participants have been asked to carry out a series of vital tasks which in a way represent the general use-cases of such wallets. We present the description of these tasks and other experimental setups in Section 4. Once completed, we have measured and analysed the rate of success and failure of completing the tasks and identified several crucial issues while using these cryptocurrency wallets. The findings of our analysis and a brief discussion are presented in Section 5. Finally, we conclude in Section 6 with a hint of the future work in this field.

2 Background

In this section, we present a brief description of the selected cryptocurrencies (Section 2.1) and their different types of wallets (Section 2.2). In addition, we present short introduction to the usability testing and its different aspects (Section 2.3).

2.1 Selected Cryptocurrencies

For this study, we have selected 3 cryptocurrencies, namely Bitcoin, Ethereum, and Ripple (a.k.a. XRP). The selected currencies represent the top-three cryptocurrencies in terms of their total market capital (a popular metric in the cryptocurrency sphere) as of 27 October, 2019 [6]. A brief introduction of the selected currencies are presented below.

- **Bitcoin** [1] is the first cryptocurrency to utilise a public blockchain system secured via a novel consensus algorithm called *Proof of Work* (PoW). It is currently the most popular virtual currency. Despite its technological breakthrough, it suffers from severe limitations such as transaction speed, issues of scalability and exorbitant energy consumption [7].
- **Ethereum** [4] is often regarded as Blockchain 2.0 as it showcases the next evolution of the blockchain technology. It facilitates the deployment and execution of computer programs, known as smart-contracts, on top of its public blockchain and thus, promotes the notion of blockchain as a platform. It also has similar limitations as Bitcoin [7].
- **Ripple** [5] is a public blockchain platform that has been created mainly to be integrated with financial institutions to facilitate instant payment, possibly in cross-border scenarios utilising its underlying protocol. It is gaining popularity because of its scalability and transaction speed.

2.2 Wallets

Bitcoin Core was the first wallet implemented by the bitcoin developers for desktop computers [8]. Since then there are several forms of wallets now. Next, we provide a brief discussion on different forms of wallets available on the market.

Desktop Wallet: Desktop wallets for cryptocurrencies are software that are available for most of the desktop operating systems such as Mac, Windows, Linux. In desktop wallets, private keys are stored in the computer. Some of

the desktop wallets are targeted for a single cryptocurrency such as Bitcoin Core while other wallets (e.g. Exodus [9]) support multiple currencies. The User Interface (UI) of Exodus wallet is shown in Figure 1.

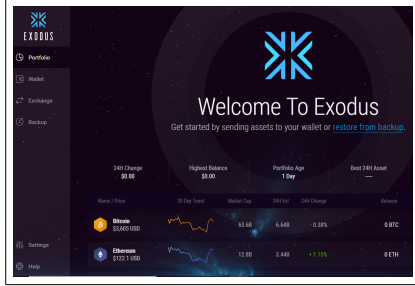


Fig. 1. User Interface of Exodus Wallet.



Fig. 2. User Interface of Jaxx Wallet.

Mobile Wallet: Mobile wallets are mobile apps (applications) whose popularities are increasing day by day with the increase of smartphone uses. Figure 2 shows the UI of Jaxx Liberty, a mobile based Multi-currency Wallet [10].

Hardware Wallet: Hardware wallets are hardware devices built specifically for handling private keys. It provides a better security option with its support for strong cryptographic mechanisms by which the keys are stored in such devices. Examples of hardware wallets are Ledger [11] and Trezor [12].

Paper Wallet: Paper wallet is simply the private keys printed on a paper, mostly used as a form of backup. As it keeps the private keys offline, it provides a way of storing and keeping cryptocurrencies safer from cyber-attacks, malware, etc. Figure 3 represents a typical paper wallet of Ethereum.



Fig. 3. Ethereum paper wallet

Web Wallet: Web wallets are internet-based wallets which can be accessed via different internet browsers, such as Google Chrome, Firefox, and so on, by visiting the websites of some specific service providers. The private keys are generally stored on the server of the respective service providers and hence, many do not consider it a safe practice to store respective keys in the server as online servers are often target for hackers. However, we have seen the emergence of some web wallets (e.g. Metamask [13]) which are browser extension or add-on and does not require any key storage in the server.

2.3 Usability & Cognitive Walkthrough

Usability is the degree to which a software is easy to use, making sure that the functionalities help a user of average ability to achieve their intended purpose [14]. J. Nielsen defined usability with the following 5 components: Learn-ability, Efficiency, Memorability, Errors and Satisfaction [15].

To evaluate the usability of a system, it is important to analyse the system, utilise the resources available and select the appropriate inspection methods, which are crucial for the desired output and greatly reduces the risk of coming out with the limited set of findings.

There are mainly two types of methods, *Analytical* and *Empirical*, which are used to evaluate any usability related issues in a system. In usability testing, an empirical method means performing a test with the real users of the product while being observed by usability experts [16]. That is, it is based on experience or direct observation rather than any quantitative analysis or pure logic. On the other hand, an analytical approach tries to simulate people using a system, typically a software or a hardware, and then uses design principles, models or guidelines by experts as participants to judge and predict likely problems. An analytical usability testing is relatively easy to perform as it only requires several usability experts to find out the most of the usability problems, thus the overhead to perform a test is minimal.

Cognitive Walk-through (C.W.) is a usability inspection method which relies on the analytical approach to carry out a usability testing. It focuses on the ease of use and learn-ability of a product from a new user's perspective, usually performed by one or more expert [17]. In cognitive walk-through, a participant evaluates the user-interface of the corresponding application and carries out a series of tasks. The tasks allow the evaluator to assess different usability aspects corresponding to the particular application. In general, a cognitive walk-through process consists of three parts:

- Pointing out the intended target.
- Identifying the tasks essential to achieve the goal.
- Evaluating the interface and reporting the usability problems.

3 Related Work

There have been a few existing works which studied the usability issues related to cryptocurrency wallets. For example, Kazerani et al. performed a study using Changetip and Coinbase to investigate what degree usability and user experience were a factor for bitcoin [18]. In their study, participants were beginner level bitcoin users and were asked to provide commentary while completing a provided task. They found that about half of the participants were unable to understand the concepts and were confused about their actions.

On the other hand, Krombholz et al. presented the first large scale survey to investigate how users experience the Bitcoin ecosystem in terms of security, privacy and anonymity [19]. They found that many users did not use all security capabilities of their selected Bitcoin management tool and had significant misunderstanding on how to remain anonymous and protect their privacy in the

bitcoin network. They also found that 22% of their participants had lost their money due to security breaches or self-induced errors.

S. Eskandari et al. first evaluated the usability and security issues in bitcoin key management and summarised the bitcoin key management approaches analysing 6 representative bitcoin clients and found that users performing tasks involving key management can be stuck with complex security issues. [20]. They also applied a cognitive walk-through inspection method along with a set of guidelines with four fundamental tasks for bitcoin but didn't showed any stats or numerical representation regarding the usability issues in bitcoin wallets.

As we can see, most usability testing for cryptocurrencies so far are based on Bitcoin. There are other significant cryptocurrencies offering a wide range of possibilities, unfortunately, there usability analysis has not been publicly reported. We aim to fill in this gap in this study where we have measured the usability of both desktop and mobile based wallets for the selected cryptocurrencies in all notable platforms applying an analytical usability inspection method, the cognitive walk-through method.

4 Experiment Setup

In this section, we present the experimental setup. In particular, we discuss about the selected wallets (Section 4.2), associated tasks (Section 4.1) and the participants (Section 4.3).

4.1 Experimental Tasks

To investigate the usability issues faced by the cryptocurrency users and to measure the effectiveness and satisfaction level, the cognitive walk-through inspection method has been selected, focusing on the essential tasks to be performed by a cryptocurrency user. The design of analytical cognitive walk-through for this study is summarised next.

For our experiment, we have selected 4 major tasks which are sub-divided into 13 sub tasks. For each sub-task, participants have been required to answer the following four standard questions which was first mentioned on the usability book by C. Wharton et al. [21] and currently is considered as standard procedures in the cognitive walk-through questionnaires. These questions essentially require each participant to answer in a binary *yes/no* fashion where a *yes* signifies that the user has been able to perform the respective task whereas a *no* signifies the user has failed.

- Q1 Will the user understand how to start the task? (understanding that the action is needed)
- Q2 Is the control(e.g. button) for the action clearly visible?
- Q3 Will the user associate the correct action with the outcome they expect to achieve? (link of the control with the action)
- Q4 If the correct action is performed, will the user see that progress is being made towards their intended outcome?

Eskandari et al.[20] in their study determined four vital tasks a cryptocurrency user needs to know while dealing with the wallets. We have reviewed those

tasks and come up with a few modifications for this study. The tasks selected for our experiment are presented below.

- T1: Configure** - Starting a wallet for the first time in a new device, creating a new account (sub-task $t1$) and checking its balance (sub-task $t2$).
- T2: Spend** - Making a transaction of a given amount of cryptocurrencies to a valid receiving address. Thus, this task involves finding the transaction functionalities (send-receive) (sub-task $t3$), entering information such as receiver's address (sub-task $t4$), amount (sub-task $t5$) and finally, submitting the transaction (sub-task $t6$).
- T3: Spend in Secondary Device** - Spending the given amount on a secondary device but from the same wallet(account) and to the same address as $T2$. This task requires taking a backup (sub-task $t7$) via a wallet from the primary device and restoring to the secondary device (sub-task $t8$) and then making the transaction (sub-tasks $t9$, $t10$ and $t11$) following the steps of Task $T2$.
- T4: Recover** - Recovering after losing the principal credential (e.g. hard drive crash). This task will require getting the right option to restore from backup to the correct state while having a backup(sub-task $t12$) or getting the verdict when the backup is not available(sub-task $t13$).

4.2 Selected Wallets

Next, in order to select the corresponding wallets, a small case study was conducted. Our goal has been to select a set of cryptocurrency wallets which covers the majority of the desktop and mobile platforms for these cryptocurrencies. The selected wallets are presented in Table 1.

Table 1. Selected Wallets for the Study

No.	Wallet Name	Platform & Type
1	Exodus	Desktop-based Multi-currency Wallet
2	Bitcoin Core	Desktop-based Bitcoin Wallet[8]
3	Jaxx	Cross Platform Multi-currency Wallet[10]
4	Ethereum Wallet	Desktop-based Ethereum Wallet
5	Coinomi	Mobile-based XRP Wallet

These wallets represent the most widely used wallets in desktop/laptop computers and mobile devices. Some of them are used as desktop wallets for a single cryptocurrency (e.g. Bitcoin Core and Ethereum Wallet for Bitcoin and Ethereum respectively) while others are multi-currency wallets, supporting different cryptocurrencies. Finally, Coinomi is a mobile-based XRP Wallet.

4.3 Participants

There were 5 participants in this study. It is widely accepted that testing with five participants is capable of finding almost as many usability problems as testing with many participants [22]. An analytical cognitive walkthrough is more favorable as it requires expert users to simulate the evaluation from the eye of a

novice user. The mean participant age in years is 30 and standard deviation is 5.98. All participants had minimum qualifications of B. Sc. in computer science with 1 year of research experience in blockchain areas and familiarity in usability. The mean experience in blockchain is 1.6 in years, standard deviation is 1.2 and each of the participants is a blockchain researcher at Shahjalal University of Science and Technology, Sylhet, Bangladesh.

Prior to the evaluation, each participant was supplied with the documentation of our designed cognitive walk-through method and other necessary tutorials. Each participant was asked to carry out the above-mentioned tasks for the selected wallets and answer 4 usability questions (Q1-Q4). Each participation lasted 80 minutes in average which was conducted in a distraction free room while all the participants evaluated the desktop wallets on a Windows-10 laptop with Intel core-i5 processor and 8GB of RAM and the mobile wallets on a Asus Zenfone 5 Android phone, with a screen recorder application running in the background during the evaluation. While carrying out the experiment, the number of *yes* and *no* for the selected tasks was recorded. At the end of the experiment, each participant also had the opportunity to provide verbal feedback.

5 Findings and Discussion

Since there were 4 significant tasks (T1-T4) broken down to 13 sub-tasks and 4 usability questions for each sub-task, an evaluator actually had submitted 52 answers in a yes-no fashion. The total number of failed steps (signified by a *no* answer) for each wallet by all the 5 participants are presented in Table 2 which also includes the average of failed steps out of 52 by all participants.

Table 2. The rate of failed steps by the participants.

Evaluator No.	Exodus	Bitcoin Core	Ethereum	Jaxx	Coinomi
E1	6	22	8	2	0
E2	26	31	9	7	5
E3	9	25	10	4	2
E4	6	22	8	2	0
E5	4	22	12	4	0
Avg. Failed Steps out of 52	10.2	24.4	9.4	3.8	1.4
% of Usability Problems	19.62%	46.92%	18.08%	7.3%	2.69%

As we can see, no wallet is without any failed steps, meaning novice users failed to perform some fundamental tasks in every wallet. To be more specific, about 47% of the steps to complete the fundamental tasks are failed on bitcoin core, which is the first appeared, one of the most popular bitcoin wallet. For other desktop wallets, the the average rate of failure are 19.62% and 18.08% respectively for Exodus and Ethereum wallet. The failure rate on the mobile wallets are 3.8% and 2.69% for Jaxx and Coinomi respectively which is much better compared to the result of desktop versions while the rate of usage of mobile based wallets are notably lower than that of desktop based wallets. The rate of success for these tasks are shown in Figure 4 along with their standard deviations.

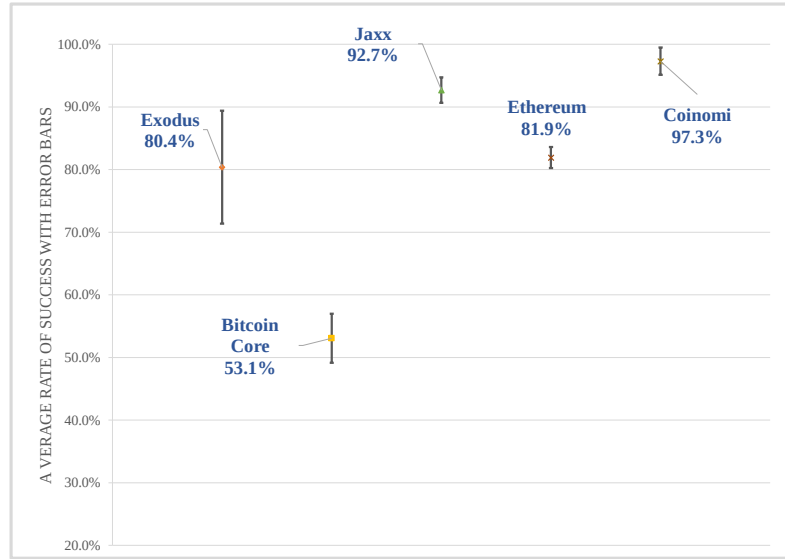


Fig. 4. The success rate of the fundamental tasks.

Figure 5 better represents the difficulties faced while completing the tasks for all the selected wallets where the bars signifies the number of failed steps out of 52 grouped by participants. It also shows the symmetry of evaluation records of the participants. The percentage of failed steps against the 4 fundamental tasks are represented in Figure 6 which shows that spending and receiving coins (T2) were the most easy task to accomplish by the participants while the remaining other tasks were difficult to complete. In particular, Task T4 (recovering a lost credential) was the most difficult task to complete. This shows that all essential tasks except spending a cryptocurrency are proven to be extremely difficult for new cryptocurrency users.

The verbal session with the participants also highlighted several aspects regarding the usability of the wallets. For example, 3 of the selected wallets generate key pairs without any knowledge of the user. This kind of abstraction is debatable though. Many users may not like to be stressed by the key generation process, however, there could be a situation where a user might want to manage those keys. The abstraction restricts a user from this option. Recovering from the key loss depends on if the user kept any backup. Even if backup sounds like an easy task, finding the right file could be a hassle.

One of the participant reported that Bitcoin core used difficult technical languages. These technical terms must be complicated to the new users. Examples of such languages in Bitcoin Core is “out of sync” or “synchronizing with network”, meaning the process of downloading the whole blockchain. The highly technical language could be replaced by simpler instructions.

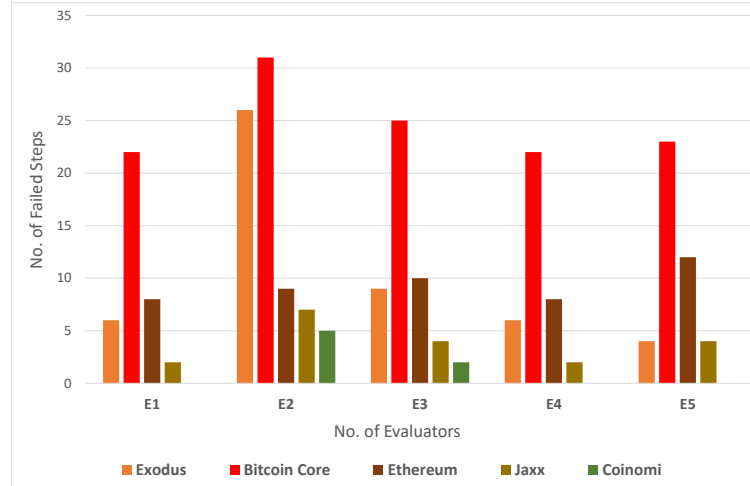


Fig. 5. The no. of failed steps by each evaluator against the selected wallets.

Ethereum wallet improved upon the difficulties of Bitcoin Core. For example, upon launching the wallet, it asks the user to create a new account or to restore an old one. It also notifies the user to keep a copy of the folder *KeyStore* for backup. The backup procedure of Ethereum wallet is also simple because it has a drop-down option of backup inside the file menu. The backup procedure maps to a folder named *KeyStore* and if that file is kept safe in a secondary device, then restoring the account to a new device is hassle-free.

As per the evaluation, Bitcoin Core wallet failed to answer the question “Will the user associate the correct action with the outcome they expect to achieve?” the highest number of times. The wallet lack interface cues and features like tool help, wizards, or any action guides. Conversely, the Ethereum wallet is more flexible in the sense that the new users can explore effortlessly.

Exodus shows the two main options, “Send Assets to Wallet” and “Restore from Backup” which is vital for any users starting to use the wallet. It comes with the simple sidebar making it easy to swap between the functionalities along with the latest price of popular cryptocurrencies. This wallet emphasises on taking offline backup and follow the state-of-art backup procedure. This was the most usable desktop based wallet in the study but still the participants failed a few sub tasks in T1, T3, and T4.

We selected two popular mobile based wallets: Coinomi and Jaxx Liberty. Mobile based wallets have more attractive interfaces with the interface cues enabling new users to get comfortable with the wallet. The backup procedure is relatively simple in such wallets. The wallets also notify the user about the risks of not taking a backup. The average rate of success in completing the fundamental tasks for Jaxx and Coinomi was 92.7% and 97.3% respectively which is pretty good compared to the desktop wallets where the success rate of Exodus, the most usable desktop wallet was around 80.4%.

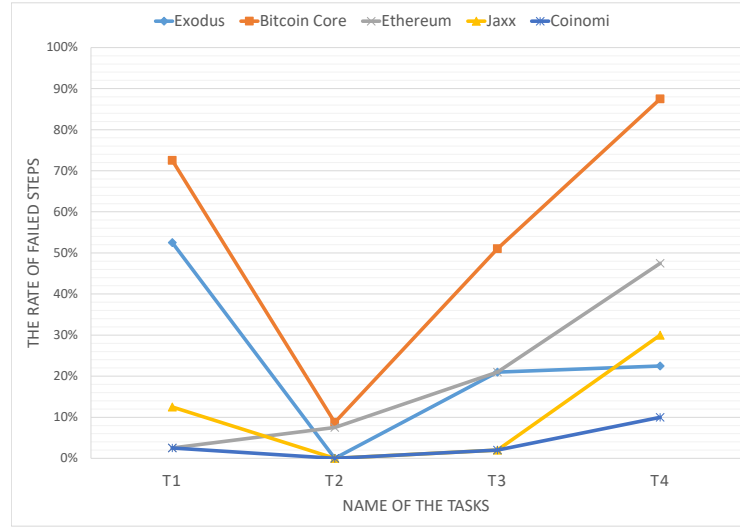


Fig. 6. The percentage of failed steps against each task for the selected Wallets

Even though the Cognitive Walkthrough is quite flexible, it has some pitfalls too. The learn-ability error bars presented as the standard deviation in Figure 4 represents that the value of standard deviation was reducing while the evaluators was gradually coming to the end of the assessment. Still, the findings on this study is clear enough to identify the usability issues in fundamental tasks in cryptocurrency wallets. Resolving these issues will significantly improve the user acceptance and global adoption of cryptocurrencies.

6 Conclusion

It is envisioned that cryptocurrencies will play a vital role in the global currency ecosystem. The more usable we can make the technology, the more practitioners, users, and system architects will be using it for business, security, and anonymity. The objective of the study presented in this paper is to analyse the usability issues of the public blockchain systems focusing on the key tasks that must be accomplished on. Cryptocurrencies are the most popular applications based on blockchain and we chose to work with the three most popular cryptocurrencies at the time, namely Bitcoin, Ethereum and Ripple.

In this paper we presented the usability issues of some currently popular wallets of these currencies focusing on a series of tasks. The wallets can be categorised as: Desktop based wallet (Bitcoin Core, Ethereum Wallet, and Exodus) and Mobile based wallet (Jaxx, Coinomi). In our evaluation, we have observed that a lot of innovative and promising initiations have been considered to solve the key management system of public blockchain system. We believe the security-usability trade-off applies to blockchains too as the key idea is to make the system secure, anonymous, and reliable. However, as our findings suggest that the software (wallets) used to access any cryptocurrency suffer from serious usability issues, in particular for general users.

A set of usability guidelines need to be developed especially for cryptocurrency wallets so as to enable the development of more usable cryptocurrency wallets and other applications. We plan to work on this in the continuation of this study in future.

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